10 Equity Analysis - Subgroup Performance and Label Sensitivity

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0.1 Introduction

Wang et al. (2022) showed that sepsis prediction models perform worse for Asian, Hispanic, and Spanish-speaking patients. This notebook extends their work with:

1. **Subgroup performance**: AUROC, AUPRC, and alert burden evaluated per demographic subgroup (race/ethnicity, language, insurance).
2. **Subgroup label-sensitivity**: How the fraction labelled septic varies across the three label variants *within each subgroup*. Non-uniform sensitivity means label choice has equity implications beyond aggregate event-rate differences.

0.2 Subgroup Performance

Whether the models discriminate equally well across demographic strata. Every metric below is computed on the same test rows and the same predictions as the aggregate results, sliced by subgroup rather than recomputed, so a difference between strata is a difference in performance and not an artefact of scoring them separately.

Two things limit what this section can conclude, and both are structural rather than incidental. Small strata carry intervals wide enough to reach the chance line, and the demographic fields themselves are administrative categories with a missingness level of their own.

0.2.1 Metric Helper

Computes AUROC, AUPRC, and alert burden for any subgroup slice with minimum-sample guards.

0.2.2 Performance by Subgroup (Variant B, Primary Model)

Subgroup levels are taken in order of sample size and scored across the primary model, GBT and NEWS2. Selecting by size keeps the choice outcome-blind, and it is also the only selection rule that leaves enough events in a stratum for the metrics to return anything at all.

Levels that clear the counting floor still need not clear the floor for being read. Stage 12 applies the second one.

Table 1: Subgroup performance (prima

	subgroup_col	subgroup_val
insurance–pred_sepsis	insurance	
insurance–pred_gbt	insurance	
insurance–pred_news2	insurance	
insurance-Medicaid-pred_sepsis	insurance	Medicaid
insurance-Medicaid-pred_gbt	insurance	Medicaid
insurance-Medicaid-pred_news2	insurance	Medicaid
insurance-Medicare-pred_sepsis	insurance	Medicare
insurance-Medicare-pred_gbt	insurance	Medicare
insurance-Medicare-pred_news2	insurance	Medicare
insurance-Other-pred_sepsis	insurance	Other
insurance-Other-pred_gbt	insurance	Other
insurance-Other-pred_news2	insurance	Other
insurance-Private-pred_sepsis	insurance	Private
insurance-Private-pred_gbt	insurance	Private
insurance-Private-pred_news2	insurance	Private
language-Chinese-pred_sepsis	language	Chinese
language-Chinese-pred_gbt	language	Chinese
language-Chinese-pred_news2	language	Chinese
language-English-pred_sepsis	language	English
language-English-pred_gbt	language	English
language-English-pred_news2	language	English
language-Haitian-pred_sepsis	language	Haitian
language-Haitian-pred_gbt	language	Haitian
language-Haitian-pred_news2	language	Haitian
language-Kabuverdianu-pred_sepsis	language	Kabuverdianu
language-Kabuverdianu-pred_gbt	language	Kabuverdianu
language-Kabuverdianu-pred_news2	language	Kabuverdianu
language-Other-pred_sepsis	language	Other
language-Other-pred_gbt	language	Other
language-Other-pred_news2	language	Other
language-Portuguese-pred_sepsis	language	Portuguese
language-Portuguese-pred_gbt	language	Portuguese
language-Portuguese-pred_news2	language	Portuguese
language-Russian-pred_sepsis	language	Russian
language-Russian-pred_gbt	language	Russian
language-Russian-pred_news2	language	Russian
language-Spanish-pred_sepsis	language	Spanish

language-Spanish-pred_gbt	language	Spanish
language-Spanish-pred_news2	language	Spanish
race-ASIAN-pred_sepsis	race	ASIAN
race-ASIAN-pred_gbt	race	ASIAN
race-ASIAN-pred_news2	race	ASIAN
race-ASIAN - CHINESE-pred_sepsis	race	ASIAN - CHINESE
race-ASIAN - CHINESE-pred_gbt	race	ASIAN - CHINESE
race-ASIAN - CHINESE-pred_news2	race	ASIAN - CHINESE
race-BLACK/AFRICAN AMERICAN-pred_sepsis	race	BLACK/AFRICAN AMERICAN
race-BLACK/AFRICAN AMERICAN-pred_gbt	race	BLACK/AFRICAN AMERICAN
race-BLACK/AFRICAN AMERICAN-pred_news2	race	BLACK/AFRICAN AMERICAN
race-HISPANIC/LATINO - PUERTO RICAN-pred_sepsis	race	HISPANIC/LATINO - PUERTO R
race-HISPANIC/LATINO - PUERTO RICAN-pred_gbt	race	HISPANIC/LATINO - PUERTO R
race-HISPANIC/LATINO - PUERTO RICAN-pred_news2	race	HISPANIC/LATINO - PUERTO R
race-OTHER-pred_sepsis	race	OTHER
race-OTHER-pred_gbt	race	OTHER
race-OTHER-pred_news2	race	OTHER
race-UNKNOWN-pred_sepsis	race	UNKNOWN
race-UNKNOWN-pred_gbt	race	UNKNOWN
race-UNKNOWN-pred_news2	race	UNKNOWN
race-WHITE-pred_sepsis	race	WHITE
race-WHITE-pred_gbt	race	WHITE
race-WHITE-pred_news2	race	WHITE
race-WHITE - OTHER EUROPEAN-pred_sepsis	race	WHITE - OTHER EUROPEAN
race-WHITE - OTHER EUROPEAN-pred_gbt	race	WHITE - OTHER EUROPEAN
race-WHITE - OTHER EUROPEAN-pred_news2	race	WHITE - OTHER EUROPEAN

0.2.3 AUROC by Race

Discrimination by racial and ethnic stratum, set beside Wang et al.'s (2022) report of degraded performance for Asian, Hispanic and Spanish-speaking patients. Bar heights on their own support no claim. The contrasts, their intervals and the false-discovery correction are computed in stage 12.

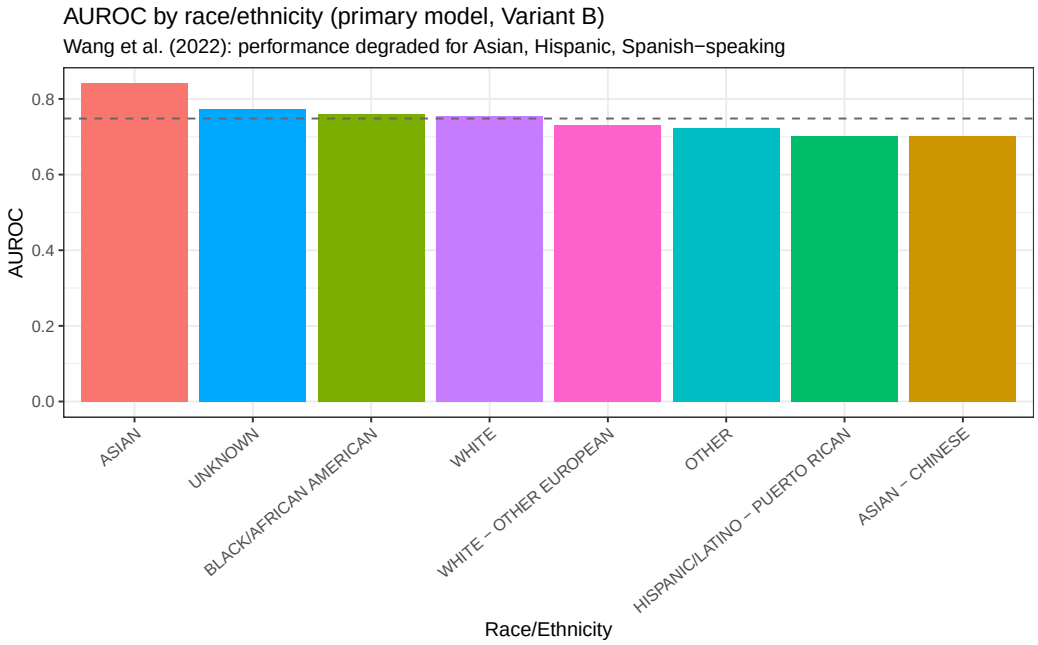


Figure 1: AUROC stratified by race/ethnicity for the primary model (Variant B). Dashed line indicates the overall mean.

0.3 Subgroup Label-Sensitivity

For each subgroup, the fraction labelled septic is computed under all three label variants. A large spread (pp range across A, B, C) means the label choice disproportionately affects that group.

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##
## Subgroups with highest label sensitivity (>5 pp range):
```

	subgroup_col	subgroup_val	pct_A	pct_B	p
race-WHITE	race	WHITE	9.4	9.6	
race-UNKNOWN	race	UNKNOWN	11.6	12.8	
race-BLACK/AFRICAN AMERICAN	race	BLACK/AFRICAN AMERICAN	9.3	9.1	
race-OTHER	race	OTHER	8.8	9.8	
race-UNABLE TO OBTAIN	race	UNABLE TO OBTAIN	9.7	11.1	
race-WHITE - OTHER EUROPEAN	race	WHITE - OTHER EUROPEAN	9.4	7.7	
race-ASIAN	race	ASIAN	10.5	11.5	
race-ASIAN - CHINESE	race	ASIAN - CHINESE	10.1	11.1	
language-English	language	English	9.5	9.8	
language-Spanish	language	Spanish	10.1	9.7	
language-Chinese	language	Chinese	11.2	11.1	
language-Russian	language	Russian	9.8	12.2	
language-Portuguese	language	Portuguese	7.5	9.8	
language-Haitian	language	Haitian	10.2	12.6	
language-Kabuverdianu	language	Kabuverdianu	10.2	10.5	
language-	language		11.6	15.3	
insurance-Medicare	insurance	Medicare	10.0	10.3	
insurance-Private	insurance	Private	8.3	8.7	
insurance-Medicaid	insurance	Medicaid	10.8	10.7	
insurance-Other	insurance	Other	10.1	10.3	
insurance-	insurance		8.5	8.7	
insurance-No charge	insurance	No charge	28.6	14.3	

0.3.1 Label Sensitivity vs. AUROC Deficit

Tests whether subgroups with higher label sensitivity also have lower AUROC. A negative trend would suggest label ambiguity compounds prediction difficulty.

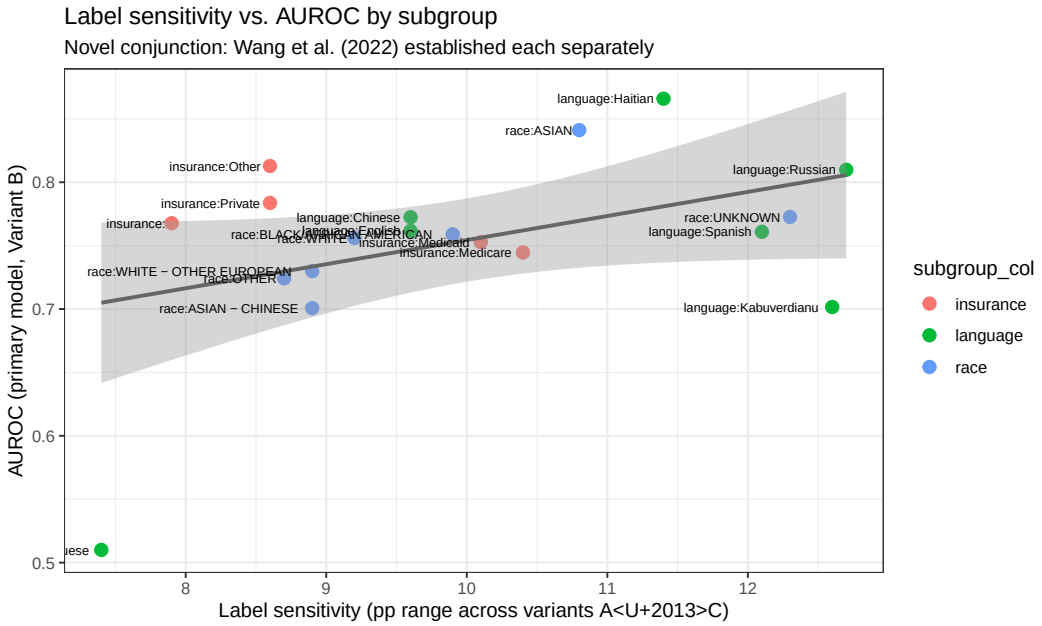


Figure 2: Subgroup label sensitivity (x) vs. AUROC (y). Each point is a demographic subgroup; a downward trend suggests label ambiguity compounds prediction difficulty.

```
##
## Equity analysis and all V2 analyses complete.
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0.4 Reading the Equity Results

Two readings are available here and only one of them is supported by this design.

The subgroup performance results are a precision problem before they are a fairness problem. Sepsis onsets are rare per stay, and splitting them across demographic strata leaves most strata with event counts that cannot carry a discrimination estimate. Where a contrast comes back with no significant difference, the honest statement is that the study could not detect one, which is not the same as showing the model treats strata alike. Stage 12 applies the interpretation floor that decides which strata can be read at all.

Label-sensitivity is on firmer ground, because it is a proportion computed over stays rather than a discrimination metric estimated from events. If the spread across variants A, B and C is uneven across subgroups, then who counts as septic depends on the definition differently for different groups, and every downstream rate inherits that. A model is not required for the effect to exist.

There is a further limit worth stating plainly: the demographic fields are administrative categories carrying their own missingness, and a stratum recording an absence of information is not a demographic answer. A comparison that clears the event floor on such a stratum still supports no claim about the group it appears to name.